Ex.NO: 9

DATE: 12/8/26

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER

AIM:

To implement the Subnetting and Configure a Subnetted network using CISCO Packet Tracer.

THEORY:

Subnetting divides a large network into smaller networks called Subnets. It improves IP address utilization network management, security and reduces network traffic.

For the network ~~192.168~~ 192.168.1.0/24, a /27 Subnet mask is used.

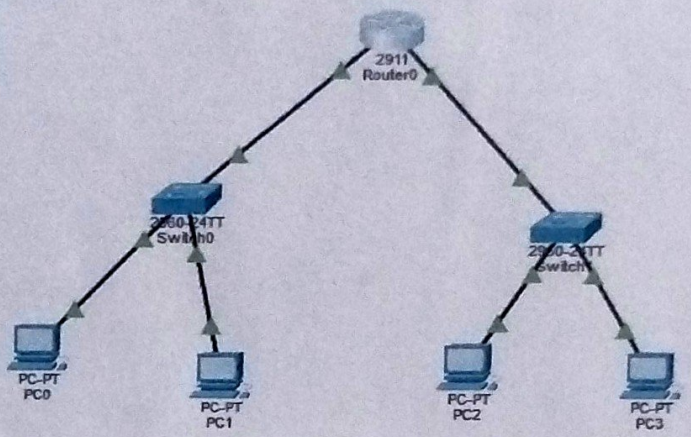
Subnet mask: 255.255.255.224

Number of Subnets: 8

Usable hosts per Subnet: 30

ALGORITHM:

1. Open CISCO Packet Tracer and Create the required network topology
2. Connect routers, Switches and PCs using appropriate cables
3. Divide the network 192.168.1.0/24 into Subnets using /27
4. Configure the router interfaces with the assigned IP addresses.
5. Configure the Switch ports in acess mode




```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=6ms TTL=128
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>
```

INPUT

Network address : 192.168.1.0/24

Number of Subnets : 8

Usable hosts per Subnet : 30

Subnet mask : 255.255.255.224 (/27)

Devices : 2 Routers, 2 switches and PCs

Ping 192.168.1.1

OUTPUT:

Reply from 192.168.1.1 : bytes = 32 time <1ms
TTL = 255

Reply from 192.168.1.1 : bytes = 32 time <1ms
TTL = 255

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TTL = 255

Reply from 192.168.1.1 : bytes = 32 time <1ms
TTL = 255

6. Assign IP address, Subnet masks and default gateway to each PC.

7. Configure routing between the routers if required.

8. Test the Connectivity using the ping command

IP ADDRESS ASSIGNMENT

Device	Interface	IP Address
Router1	G0/0	192.168.1.1
Router2	G0/1	192.168.2.1
Router2	F0/0	192.168.3.1
Router2	F0/1	192.168.4.1
PC1	—	192.168.1.11
PC2	—	192.168.1.12
PC3	—	192.168.1.13
PC4	—	192.168.1.14
PC5	—	192.168.1.15

COMMANDS FOR ROUTERS

ROUTER 1

enable
configure terminal

interface gigabitethernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit

interface gigabitethernet 0/1
ip address 192.168.2.1 255.255.255.224
no shutdown
exit

ROUTER 2

enable
configure terminal

interface fastethernet 0/0
ip address 192.168.3.1 255.255.255.224
no shutdown
exit

interface fastethernet 0/1
ip address 192.168.3.1 255.255.255.224
no shutdown
exit

SWITCH CONFIGURATION

```
enable
configure terminal
interface fastethernet 0/1
switchport mode access
exit
interface fastethernet 0/2
switchport mode access
exit
```

PC CONFIGURATION

For each PC configure:

IP Address: 192.168.x.11
Subnet Mask: 255.255.255.224
Default Gateway: 192.168.x.1
Assign different host addresses (.11, .12, .13, .14, .15) to the PCs according to their Subnet

RESULT:

Therefore a Simple topology with one configured router two Switches and PCs using Cisco Packet tracer is done.